Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	15 February 2025
Team ID	LTVIP2026TMIDS54144
Project Name	ShopSmart – Digital Grocery Web Application
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	User Registration & Login	USN-1	User Registration	2	High	Sai
Sprint-1	User Registration & Login	USN-2	Email Confirmation	1	High	Gamya
Sprint-1	User Registration & Login	USN-3	Gmail Registration	2	Medium	Siva Kumar
Sprint-1	User Registration & Login	USN-4	Login with Email & Password	1	High	Sowjanya
Sprint-1	User Registration & Login	USN-5	JWT Authentication Setup	3	High	Sai
Sprint-1	User Registration & Login	USN-6	Role-Based Middleware	3	High	Sai
Sprint-2	Product Management	USN-7	Add Product (Admin)	3	High	Gamya
Sprint-2	Product Management	USN-8	Update Product	3	High	Gamya
Sprint-2	Product Management	USN-9	Delete Product	2	Medium	Gamya

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2	Product Management	USN-10	View Products	2	High	Siva Kumar
Sprint-2	Cart Module	USN-11	Add to Cart	3	High	Sowjanya
Sprint-2	Cart Module	USN-12	Remove from Cart	2	Medium	Sowjanya
Sprint-2	Cart Module	USN-13	Update Quantity	2	Medium	Sowjanya
Sprint-3	Order Management	USN-14	Place Order	3	High	Sai
Sprint-3	Order Management	USN-15	Order History	2	Medium	Siva Kumar
Sprint-3	Order Management	USN-16	Order Tracking	3	Medium	Gamya
Sprint-3	Payment Integration	USN-17	Stripe Integration	5	High	Sai
Sprint-3	Payment Integration	USN-18	COD Option	2	Medium	Sowjanya
Sprint-3	Admin Dashboard	USN-19	View Orders	3	High	Gamya
Sprint-3	Admin Dashboard	USN-20	Generate Sales Report	3	Medium	Sai

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	12	6 Days	16 Feb 2025	21 Feb 2025	12	21 Feb 2025
Sprint-2	17	6 Days	23 Feb 2025	28 Feb 2025	17	28 Feb 2025
Sprint-3	21	6 Days	02 Mar 2025	07 Mar 2025	21	07 Mar 2025

A burn down chart is a graphical representation of work left to do versus time. Below is the burndown data for Sprint-1 (6 Days duration, starting with 12 Story Points):

Day 1 – 12 Points Remaining
Day 2 – 10 Points Remaining

Day 3 – 8 Points Remaining
Day 4 – 5 Points Remaining
Day 5 – 2 Points Remaining
Day 6 – 0 Points Remaining

This data shows a steady, consistent completion of tasks across the sprint, matching the ideal burndown trajectory for agile project management.

Velocity:
Total Story Points = 12 (Sprint 1) + 17 (Sprint 2) + 21 (Sprint 3) = 50 Points
Number of Sprints = 3

Velocity Calculation: Total Story Points / Number of Sprints
Velocity = 50 / 3 = 16.6

Average Velocity ≈ 17 Story Points per Sprint. This indicates that the team can confidently commit to and complete approximately 17 story points of work within a standard 6-day sprint cycle.

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

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https://www.visual-paradigm.com/scrum/scrum-burndown-chart/

https://www.atlassian.com/agile/tutorials/burndown-charts

Reference:

https://www.atlassian.com/agile/project-management

https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software

https://www.atlassian.com/agile/tutorials/epics

https://www.atlassian.com/agile/tutorials/sprints

https://www.atlassian.com/agile/project-management/estimation

https://www.atlassian.com/agile/tutorials/burndown-charts